Phenotyping in the OMOP Common Data Model

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Preface

Is this book for me?

This book is part of the collection of **Tidy R programming in OMOP** see accompanying books:

(1) Tidy R programming with the OMOP common data model
(2) Tidy R packages with the OMOP common data model
(3) Drug Utilisation Studies in the OMOP Common Data Model
(4) Phenotyping in the OMOP Common Data Model

We've written this book for anyone interested to conduct studies using the OMOP Common Data Model and working with databases using a tidyverse style approach. That is, human centered, consistent, composable, and inclusive (see https://design.tidyverse.org/unifying.html for more details on these principles).

New to R? We recommend you compliment the book with R for data science

New to databases? We recommend you take a look at some web tutorials on SQL, such as SQLBolt or SQLZoo

New to the OMOP CDM? We'd recommend you pare this book with The Book of OHDSI

New to the Tidy R in OMOP ecosystem? Probably you should start with the book Tidy R programming with the OMOP common data model before reading this one.

Package	Version	Link
CodelistGenerator	4.0.2	
PhenotypeR	0.5.0	
duckdb	1.5.2	
omock	0.6.2	

How is the book organised?

TO ADD

Citation

TO ADD

License

Code

The source code for the book can be found at this Github repository

Packages version

This book is rendered using the following version of packages:

```
Finding R package dependencies ... Done!
```

Note we only included the packages called explicitly in the book.

1. Proposed structure for the book

- Preface
- Introduction
  - What is Phenotyping
  - The phenotyping process
- Database Diagnostics
  - Introduction
  - Metadata
  - The person table
  - The observation period
  - Clinical tables
  - Record trends
- OMOP CDM vocabularies
  - Introduction
  - Condition based concepts
  - Drug based concepts
  - Measurement based concepts
  - Observation based concepts
  - Procedures based concepts
  - Devices based concepts
- Codes in OMOP CDM
  - Introduction
  - Codelists

  - Codelist with details
  - Concept set expression

- Create codelise
  - Introduction
  - get candidate codelists
  - Create a condition based codelist (example stroke)
  - Create a
  - Create a drug based codelist (example chemotheraphy)
  - Create a drug based codelist (example)
- Codelist diagnostics
  - Code use
  - Orphan codes
  - Drug exposure diagnostics
  - Measurement diagnostics
- Create cohorts
  - Base cohorts
  - Demographic cohorts
  - Inclusion criteria
  - Set the end date
  - Combine cohorts
  - Example cohorts (obesity + causal inference centered on a complicated end date e.g. venla study)
- Cohort diagnostics
  - Counts and attrition
  - Characterisation and large scale characterisation
  - Matched comparison
  - Overlap and timing
  - Survival
- Population diagnostics

  - Incidence
  - Prevalence

- Study workflow and final remarks

```
library(CodelistGenerator)
library(PhenotypeR)
library(omock)
library(duckdb)
```

```
Loading required package: DBI
```

Part I.

introduction/index.qmd

2. Vocabularies

Final remarks

Drug Utilisation Studies in the OMOP Common Data Model aims to ...

Learning more

If you find this book useful then joining the **Tidy R in OMOP** OHDSI working group will likely be of interest. Building on many of the concepts and tools seen in this book, the Tidy R in OMOP OHDSI working group aims to (1) create and promote a unified set of resources to guide Tidy R development in OMOP and support the OHDSI community, and (2) establish an overview of available packages relevant to Tidy R programming in OMOP (tidyverse style packages). If you are interested in joining the working group then please email Martí Català or Raivo Kolde.

Support us

We encourage you to support this work by either citing the book in your papers or documentation, recommending it to your colleagues, or starring the GitHub repository, or simply letting us know how it helped you. Most importantly, please **use it** in research that results in patient benefit.